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Exercise Sheet 5

Exercise 1 — *Structure factor of an infinite linear atomic chain* [moderate]

- (a) Draw the reciprocal lattice of a uniform chain of Cu atoms with lattice constant a .
- (b) Draw the reciprocal lattice if additional atoms (Cu, Zn, or Au) are placed at the center between two atoms.
- (c) Calculate the structure factor $S_{\mathbf{G}}$ (assume the atomic form factor $f \sim Z$) for the chains of (a) and (b).
- (d) Start from the uniform chain and replace every 4th atom by Zn or by Au. Calculate the new $S_{\mathbf{G}}$.

Exercise 2 — *Diffraction from a simple cubic lattice with a two-atom basis* [moderate]

Consider a simple cubic crystal with lattice constant a and a two-atom basis at the positions

$$\mathbf{r}_1 = \delta a \hat{e}_x, \quad \mathbf{r}_2 = -\delta a \hat{e}_x, \quad (\delta < \tfrac{1}{4}).$$

Assume that the atomic form factors are f_1 and f_2 for the two atoms, respectively.

- (a) Derive the structure factor S_{hkl} and the corresponding intensities I_{hkl} and $I_{\bar{h}kl}$ for the case of *identical* atoms ($f_1 = f_2 = f$).
- (b) Repeat the calculation for the case of *different* atoms ($f_1 \neq f_2$). Discuss under what conditions certain reflections vanish.
- (c) For identical atoms and $\delta = \frac{1}{4}$, what happens to the structure factor S_{hkl} for odd h ?

Exercise 3 — *Cubic perovskite BaTiO_3 investigated by X-ray diffraction* [moderate]

The cubic perovskite structure of BaTiO_3 can be described by a simple cubic unit cell with:

- Ba atoms located at the cube corners,
- one Ti atom at the body center,
- and O atoms at the centers of the cube faces.

This structure can be viewed as a simple cubic lattice with a multi-atomic basis.

- (a) Describe the structure with a suitable lattice and basis.
- (b) Determine the intensity relationship between the first four Bragg reflections using the structure factor. Take $f_{\text{Ba}} = 7f_{\text{O}}$ and $f_{\text{Ti}} = 3f_{\text{O}}$ and $I \propto |S_{hkl}|^2$.